Shiheng Zhang

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Google Scholar | GitHub

Research Interests

Numerical analysis and scientific computing. Dynamical low-rank approximation and model reduction for high-dimensional and multiscale PDEs; stable and structure-preserving time integration; scientific machine learning and optimization; approximation and error analysis of value functions.

Education

Purdue University, West Lafayette, IN 09/2019 – 12/2023

Ph.D. in Computational and Applied Mathematics
Advisor: Jie Shen
Thesis: Energy-dissipative methods in numerical analysis, optimization and deep neural networks for gradient flows and Wasserstein gradient flows

Southern University of Science and Technology, Shenzhen, China 09/2015 – 06/2019

B.S. in Computational and Applied Mathematics
Advisors: Jiang Yang and Bingsheng He

Research Experience

University of Washington, Seattle, WA 01/2024 – Present

Department of Applied Mathematics
Pearson Fellow and Acting Instructor; research with Jingwei Hu

- Dynamical low-rank algorithms for kinetic equations and related PDEs.
- Structure-preserving schemes and scientific machine learning for large-scale computation.
- Error analysis of compressed value functions and their use in planning.

Awards and Honors

- Boeing Research Award, University of Washington, 2025
- T.T. Moh Graduate Scholarship, Purdue University, 2019
- Summa Cum Laude Graduate, Southern University of Science and Technology, 2019
- First Prize, Chinese Mathematics Competitions (Guangdong), Chinese Mathematical Society, 2017

Publications

Full publication list: Google Scholar

Conference Papers

C1. **Zhang, S.** *Ordinal geometry complements reconstruction: diagnosing planning with compressed value functions.* **Advances in Neural Information Processing Systems (NeurIPS 2026).** (Code).

Journal Articles

J1. Zhang, J., **Zhang, S.** & Lin, G. *Low-rank evolutionary deep neural networks via adaptive tangent-space reduction.* **Computer Methods in Applied Mechanics and Engineering** 464, 119504 (2027).

J2. **Zhang, S.** & Hu, J. *On the stability of the low-rank projector-splitting integrator for hyperbolic and parabolic equations.* **Journal of Scientific Computing** 109, 47 (2026).

J3. **Zhang, S.** & Hu, J. *A separable and asymptotic-preserving dynamical low-rank method for the Vlasov–Poisson–Fokker–Planck system.* **ESAIM: M2AN** 60, 1715–1739 (2026).

J4. **Zhang, S.,** Shen, J. & Hu, J. *SAV-based entropy-dissipative schemes for a class of kinetic equations.* **SIAM Journal on Scientific Computing** (2025).

J5. Einkemmer, L., Hu, J. & **Zhang, S.** *Asymptotic-preserving dynamical low-rank method for the stiff nonlinear Boltzmann equation.* **Journal of Computational Physics** (2025).

J6. **Zhang, S.** & Shen, J. *Structure preserving schemes for a class of Wasserstein gradient flows.* **Communications on Applied Mathematics and Computation** (2025).

J7. **Zhang, S.**[*]**,** Zhang, J.[*], Shen, J. & Lin, G. *A relaxed vector auxiliary variable algorithm for unconstrained optimization problems.* **SIAM Journal on Scientific Computing** (2025). ([*]Equal contribution).

J8. Zhang, J.[*], **Zhang, S.**[*]**,** Shen, J. & Lin, G. *Energy-dissipative evolutionary deep operator neural networks.* **Journal of Computational Physics** (2024). ([*]Equal contribution).

Preprints

R1. **Zhang, S.** *Robust high-order BUG integrators without basis augmentation.* arXiv: 2611.15175.

R2. **Zhang, S.** & Hu, J. *A reduced-order model for the high-field Vlasov–Poisson–Fokker–Planck system based on a micro–macro decomposition.* arXiv: 2610.25569. Submitted.

R3. **Zhang, S.** & Hu, J. *Robust high-order projector-splitting integrators.* arXiv: 2608 . 17157. Submitted.

R4. **Zhang, S.** *A counterexample to robust second-order convergence of the Strang projector-splitting integrator.* arXiv: 2608.12893. Submitted.

R5. **Zhang, S.** & Shen, J. *SAV schemes with decomposition-induced pullback corrections for gradient flows.* arXiv: 2606.18551. Submitted.

R6. **Zhang, S.** *Planner-admissible graph-PDE value extensions for sparse goal-conditioned planning.* arXiv: 2605.19185. Submitted.

Research Software

Ordinal geometry: evaluating compressed value functions GitHub

Python tools for evaluating action rankings and planning with compressed value functions, with experiment summaries and a CPU demo for the NeurIPS 2026 paper.

DLRA: dynamical low-rank Boltzmann solver GitHub

Python implementations of full and dynamical low-rank solvers for the stiff nonlinear Boltzmann equation, with a command-line interface, numerical tests, and simulation examples.

DLR-VPFP: low-rank kinetic simulation GitHub

First- and second-order JAX low-rank IMEX solvers for the Vlasov–Poisson–Fokker–Planck system, with a full-tensor reference solver and reproducible experiments.

RVAV: structure-preserving optimization GitHub

PyTorch implementation of the relaxed vector auxiliary variable optimizer and its momentum variant, with an optimizer interface and reproducible optimization examples.

VPFP inverse demo: differentiable kinetic simulation GitHub

JAX parameter estimation for a low-rank kinetic solver, with gradient checks, refined full-tensor references, noisy observations, and held-out-time validation.

Teaching

Instructor, University of Washington

- AMATH 351 A: Introduction to Differential Equations and Applications – Autumn 2026
- AMATH 351 A and B: Introduction to Differential Equations and Applications – Spring 2026
- AMATH 352 B: Applied Linear Algebra and Numerical Analysis – Winter 2026
- AMATH 351 A: Introduction to Differential Equations and Applications – Spring 2025
- AMATH 352 B: Applied Linear Algebra and Numerical Analysis – Winter 2025
- AMATH 351 A: Introduction to Differential Equations and Applications – Spring 2024

Teaching Assistant, Purdue University

Courses: MA 265: Linear Algebra; MA 266: Ordinary Differential Equations; MA 261: Multivariate Calculus; MA 162: Calculus II; MA 161: Calculus I

Teaching Development

AMATH 351 Lab: interactive differential equations experiments Autumn 2026

Online lab | GitHub

- Designed and developed browser-based experiments for AMATH 351, connecting direction fields, initial conditions, existence and uniqueness, and analytical solution methods through interactive visualizations.
- Built prediction prompts, parameter exploration, and solution comparisons into the activities, with mathematical explanations, keyboard-accessible controls, and automated checks of the mathematical models.

Talks and Posters

Talks

T1. **Zhang, S.** *Robust projector-splitting Runge–Kutta integrators of orders two and three.* Minisymposium talk, SIAM Conference on Computational Science and Engineering (CSE27), Pittsburgh, PA. 2027.

T2. **Zhang, S.** *On the stability of the low-rank projector-splitting integrator for parabolic equations.* Talk, CHaRMNET Annual Meeting, University of Washington. 2025.

T3. **Zhang, S.** *Structure preserving schemes for a class of Wasserstein gradient flows.* Invited minisymposium talk, SIAM Annual Meeting, Spokane, WA. 2024.

T4. **Zhang, S.** *A relaxed vector auxiliary variable algorithm for unconstrained optimization problems.* Invited minisymposium talk, SIAM Conference on Optimization, Seattle, WA. 2023.

Posters

P1. **Zhang, S.** *A separable and asymptotic-preserving dynamical low-rank method for the Vlasov–Poisson–Fokker–Planck system.* Poster, IPAM, UCLA, Los Angeles, CA. 2026.

P2. **Zhang, S.** *SAV-based entropy-dissipative schemes for a class of kinetic equations.* Poster, ICERM, Providence, RI. 2025.

P3. **Zhang, S.** *Asymptotic-preserving dynamical low-rank method for the stiff nonlinear Boltzmann equation.* Poster, CHaRMNET Annual Meeting, Virginia Tech. 2024.

P4. **Zhang, S.** *SAV-based entropy-dissipative schemes for a class of kinetic equations.* Poster, NSF Computational Mathematics PI Meeting, University of Washington. 2024.

Conference Organization

Co-organizer, Minisymposium: *Low-Rank and Tensor Methods for High-Dimensional PDEs*, SIAM Conference on Computational Science and Engineering (CSE27), Pittsburgh, PA, February 2027

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